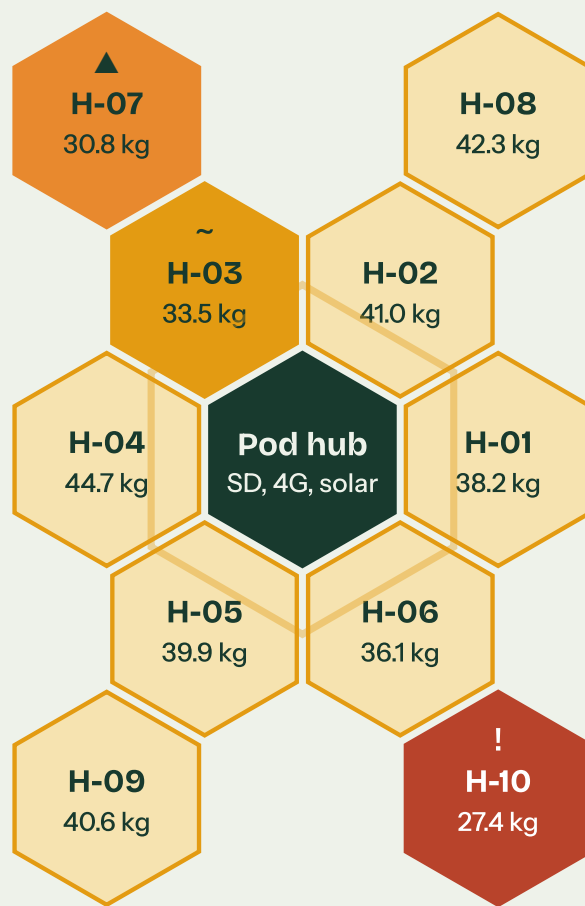


Honey Chain, built for the apiary, the field and the shopping cart

Ten hives, one weatherproof box, and no signal required. This is the fuller blueprint: what the hardware looks like, how the software keeps working offline, how an online shop can sell the honey, and what the pod costs against what ten hives yield.

<u>The pod</u>	One hub and ten hive nodes. Roughly ₹26,000 to ₹42,000 to build, or ₹13,000 to ₹22,000 in a lighter version that weighs only three hives.
<u>Offline-first</u>	Hives log to an SD card, phones keep working without signal, and everything syncs and anchors on the chain later.
<u>E-commerce</u>	Each sealed batch becomes a verified listing with a QR code, a harvest story and a lab report attached.
<u>Cost and yield</u>	Ten hives give about 100 to 500 kg a year. The pod pays for itself only if the honey sells above commodity price.



◻ Healthy
 ◻ ~ Moderate
 ◻ ▲ Warning
◻ ! Critical

The dashboard a beekeeper sees for one pod. Each cell is a hive; weights are sample data.

What changes from the first blueprint

The original plan proves the loop from hive reading to consumer scan. This version keeps that loop and adds the four things a real pilot will run into: hardware that scales past one hive, patchy connectivity, a route to market, and a budget.

Area	First blueprint	Extended blueprint
Hardware	One ESP32 prototype with temperature, humidity and weight sensing.	An apiary pod: one solar-powered hub plus ten hive nodes. A lighter “sentinel” version weighs three hives and probes the rest.

Area	First blueprint	Extended blueprint
Connectivity	Assumes the network is there.	Offline-first: SD-card logging, local alarms, SMS alerts, a phone app that works without signal, and deferred sync and anchoring.
Trust	Batch hash stored on a blockchain.	Hash-chained logs, anchored chain heads, signed QR codes, lab reports and tamper-evident seals. Plus a plain statement of what the chain can and cannot prove.
Market	Verified batch listings.	Auto-filled shop listings, pre-orders from the harvest forecast, reorder by hive, and connectors for marketplaces and bulk buyers.
Economics	Not covered.	Parts list, running costs, cost per kilogram of honey, and a break-even test.

Architecture: five layers instead of four

Two layers are new. The edge layer is what lets the system work in a field with no signal, and the commerce layer is what turns verified batches into income.

Smart hive layer	Ten hive nodes report weight, in-hive temperature and humidity. One hub collects everything, keeps time, and talks to the outside world. ESP32-C3 nodes, HX711 load cell amplifiers, SHT31 probes, ESP-NOW radio to the hub, optional microphone, camera and GPS.
Edge and sync layer	The hub writes every reading to an SD card as a hash-chained log, runs simple health rules on the spot, and raises alarms locally. It queues uploads until a connection appears. SD log, real-time clock, buzzer and LED, SMS through the 4G modem, Bluetooth and Wi-Fi-direct local dashboard, MQTT store-and-forward.

AI intelligence layer

Cloud models score health, spot anomalies and forecast the harvest. Small rule sets and a tiny model run on the hub so alerts do not depend on the cloud.

Random Forest or XGBoost, Isolation Forest, image models for later, TensorFlow Lite Micro on the hub.

Trust and traceability layer

Batches get device-generated IDs. Their hashes, and the head of each hive's log chain, are anchored on the blockchain. QR codes carry a signed summary.

Solidity contract on an Ethereum-compatible network, Web3.py, PostgreSQL off-chain, lab reports, tamper-evident seals.

Ecosystem and commerce layer

Extended

Beekeeper, admin, consumer and now buyer interfaces. A storefront and marketplace connectors publish listings straight from sealed batches and take orders back into the batch record.

React PWA, FastAPI, storefront widget, catalogue export for marketplaces and ONDC, subscriptions, recall by batch ID.

The chain of custody, with offline in mind

Stages that can happen with no signal are tagged. Only anchoring and publishing need a connection, and both can wait.

In the field

Hive to sealed batch.

1

Hive readings

Nodes report to the hub every 10 minutes.

Works offline

2

Health and alerts

Rules on the hub raise a light, a buzzer or an SMS.

Works offline

3

Harvest logged

A steep, sustained weight drop confirms it. The beekeeper adds the quantity.

Works offline

4

Batch sealed

The phone creates the batch ID and locks the record.

Works offline

Lab and ledger

Where trust is added.

5

Lab report attached

Moisture, HMF, sugar profile and adulteration screens, linked by batch ID.

6

Hash anchored

Batch hash and log-chain head go on chain when signal returns.

Can wait

7

Jars packed

QR label and tamper-evident seal applied. QR carries a signed summary.

Works offline

Shop and shelf

From batch to buyer.

8

Listing published

Origin, harvest date, hive curve and lab report fill the page automatically.

9

Order and dispatch

Each stage is logged against the batch.

10

Consumer scan

The verification page shows source, timeline and chain status.

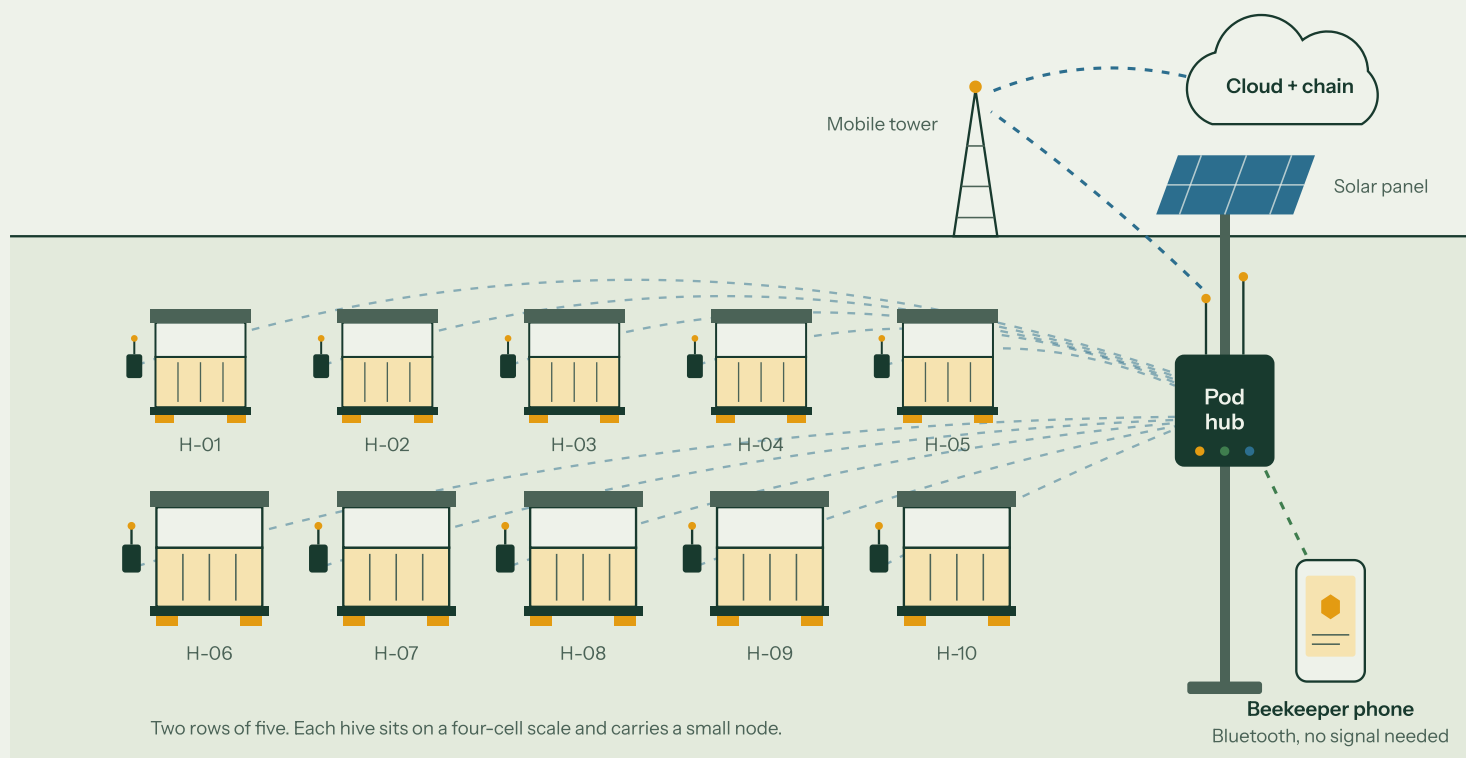
11

Feedback returns

Reorders and ratings flow back to the beekeeper by batch and hive.

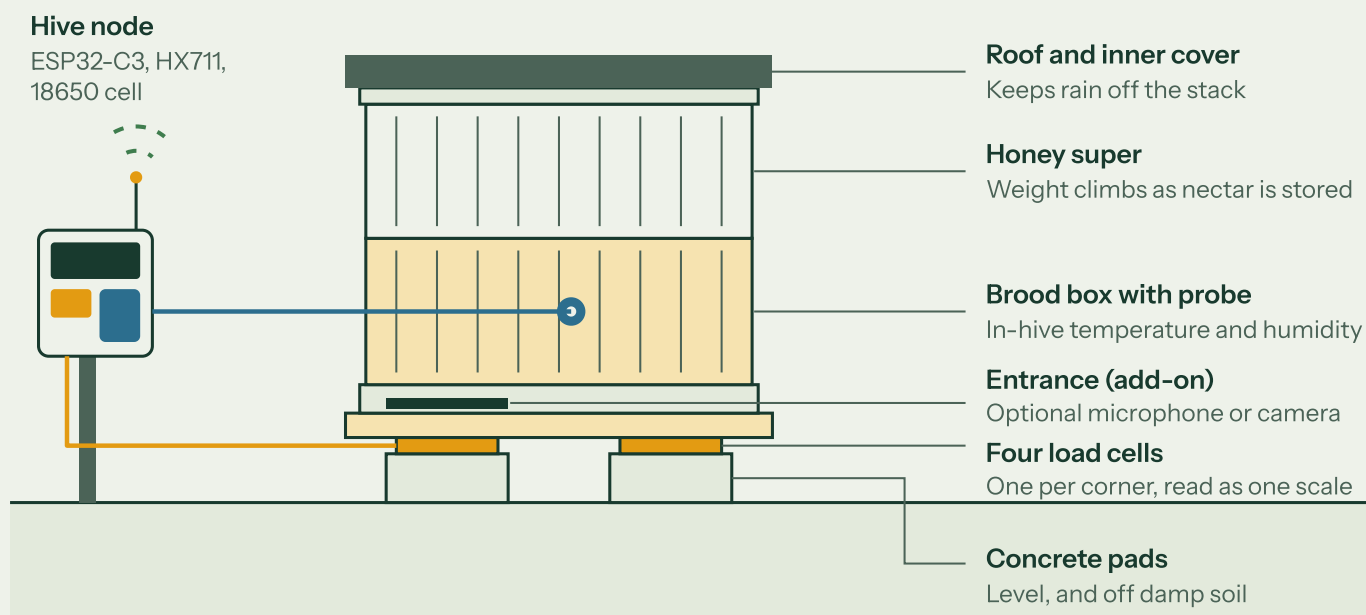
The apiary pod: one box for ten hives

A pod is one hub on a pole plus ten small hive nodes. Nodes wake every ten minutes, read their sensors, send one short radio packet to the hub and go back to sleep. The hub is the only part that needs a big battery, a solar panel and a SIM.



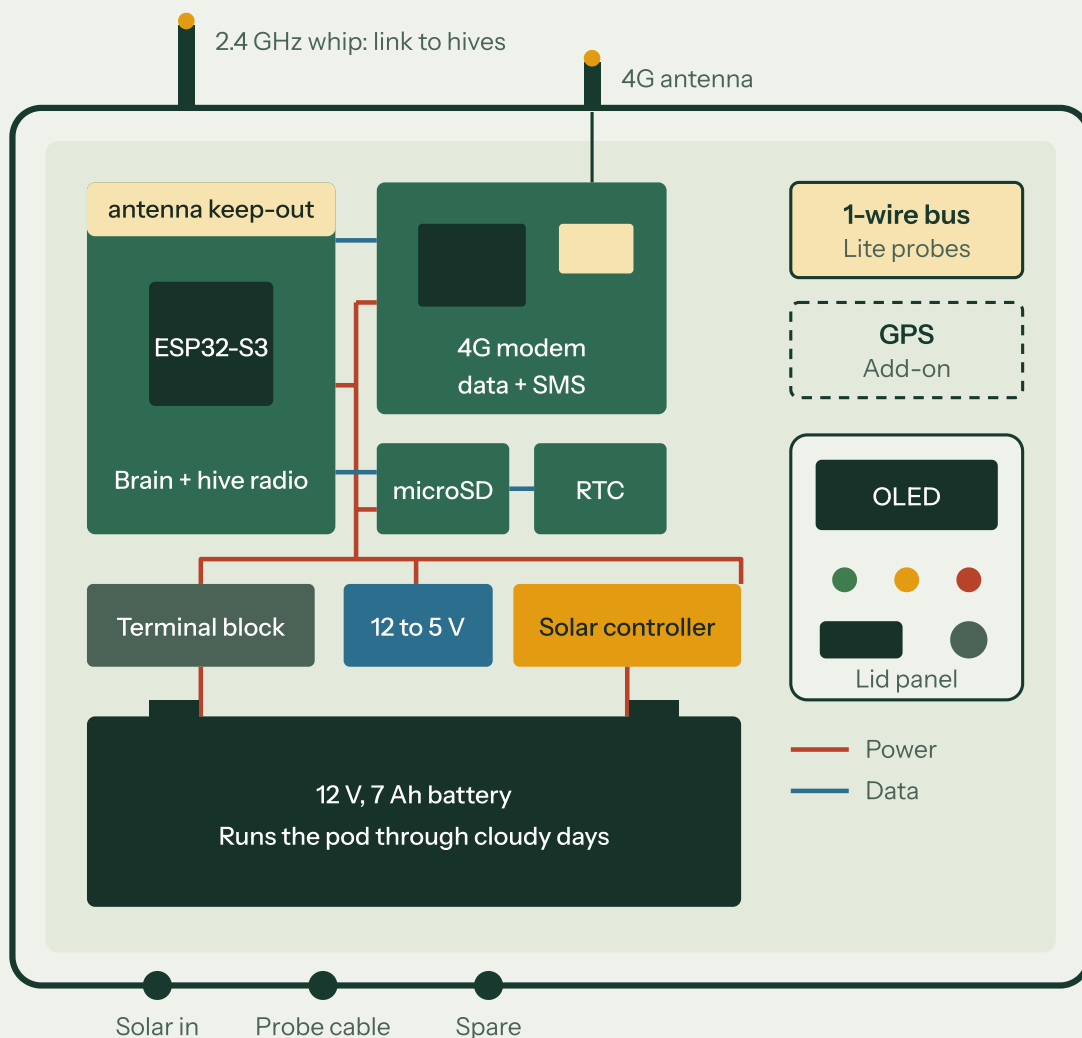
Scroll sideways to see the whole figure.

Top-level view of a standard pod. Dashed lines are radio links to the hub; the hub reaches the cloud through the mobile network, and reaches the beekeeper's phone directly over Bluetooth when there is no signal.



Scroll sideways to see the whole figure.

One hive node. The scale is four small bar-type load cells wired together; the probe sits in the brood box, where temperature matters most.



Scroll sideways to see the whole figure.

Inside the hub. It stays awake, listens for all ten nodes, and logs to the SD card first, then uploads when it can.

Two ways to build it

Standard pod. Every hive gets a full node: scale, temperature and humidity. Best when the beekeeper sells premium honey and wants hive-level stories for every jar.

Ten nodes with ESP32-C3, HX711, four load cells and an SHT31 probe, plus the hub.

Lite pod, with sentinel hives. Three hives get full nodes. The other seven get a wired temperature probe on a cable to the hub. The three scales tell you when the whole apiary is in a nectar flow.

Cheaper, but only three hives have their own weight curve. Choose sentinels that represent weak, average and strong colonies.

Why nodes are wireless and the hub is not. Long cables between ten hives collect noise and get chewed. A node that sleeps between readings runs on one 18650 cell for

months, so only the hub needs solar. Battery life depends on radio range and sampling interval; measure it on your own hives before promising anything.

Offline-first: the field decides, the cloud catches up

Most apiaries sit where mobile data is weak or absent. So every step a beekeeper needs on the day works locally, and the cloud is where records are backed up, checked and anchored, not where they are created.

Hive and hub

- Hash-chained log on SD card
- Clock from RTC, corrected by network or GPS time
- Rules for temperature, humidity, weight drops and silent nodes
- Buzzer, LED and SMS alerts

ESP-NOW radio, or a cable

Beekeeper phone

- Progressive web app or React Native app
- Local database with an outbox queue
- Reads live data from the hub over Bluetooth or Wi-Fi direct
- Registers hives, logs inspections, seals batches

4G or Wi-Fi, whenever it appears

Cloud

- PostgreSQL and the full AI models
- Accepts each record once, so retries are safe
- Anchoring service writes to the chain

- Public verification pages and shop connectors

Every record moves through three states

Saved on device

Recorded on the hub or phone, with a device ID, a counter and the hash of the previous record.
Visible immediately to the beekeeper.

Synced to cloud

Backed up and checked for gaps. Visible to admins and buyers with access. The record is safe but not yet independently timestamped.

Anchored on chain

The batch hash and the log-chain head are on the blockchain. Anything earlier in the chain can no longer be quietly changed. This is the state consumers see as “verified”.

What still works with no signal

Task	With no network	When the network returns
Live hive readings	Hub display and the beekeeper's phone, over Bluetooth or Wi-Fi direct.	Backlog uploads in bulk, oldest first.
Alerts	Local buzzer and LED. SMS through the 4G modem, which needs far less signal than mobile data.	Alert history syncs to the dashboard.
Health score	Simple rules and a small model on the hub and phone.	Cloud models re-score the backlog.
Hive registration, inspections, harvest	Saved in the phone's database; IDs are made on the device.	Outbox is sent; the server accepts each record once.
Batch creation	Sealed on the phone. Batch ID is cluster code, device code and a counter, so two phones never collide.	Hash anchored on chain.

Task	With no network	When the network returns
QR label	Generated locally. It carries a short signed summary of the batch.	Public page fills in chain status and full timeline.
Consumer scan	A cached web app can check the QR signature and show the saved summary, and says clearly that live status needs signal.	Full verification page loads.

Conflict and tamper rules

- Sensor readings are append-only, so they cannot conflict.
- Sealed batches never change. A correction creates a new version linked to the old one.
- A missing counter or a broken hash link shows up as a gap, not a silent edit.
- Retried uploads are harmless because the server keys every record by device ID and counter.

Being honest about time

- An offline record can be back-dated until it is anchored. So the interface says “recorded” until anchoring, then “anchored”.
- The hub's clock is the only witness in the field. Use a real-time clock with a coin cell, and correct it whenever network or GPS time is available.
- Anchor often. A daily chain-head anchor per pod limits how much can be disputed.

Language and literacy. Build the beekeeper app icon-first, in Hindi and regional languages, with voice prompts for the three actions used most: check hives, log harvest, seal batch. This moves from “future scope” to a v1 requirement, because a rural beekeeper who cannot read the screen will not use the system.

How an online marketplace sees a Honey Chain seller

A marketplace has one big problem with honey: buyers cannot tell real from adulterated, so honest sellers compete with cheap fakes. Honey Chain gives the marketplace evidence it can show, and gives the beekeeper a reason to be paid more.

What the marketplace worries about	What Honey Chain hands over
Is this honey adulterated?	A lab report linked to the batch hash, a sealed batch, and random re-tests of stock from the same batch.
Is the seller real and licensed?	A beekeeper registry with the seller's licence and food-safety registration on file. Identity checks happen off chain.
Will the stock be there?	A harvest forecast from hive weight and the production model, so listings can say “expected in three weeks” and take pre-orders.
Will customers return it or dispute it?	A verified-batch page that answers “is this real?” before the customer asks, and batch-level recall if a problem is found.
Can the product tell a story that converts?	A hive-level timeline: flow season curve, harvest date, declared region, and the beekeeper's name.

A listing, generated from the batch



Forest wildflower honey, 500 g
₹649

Verified batch

Batch HC-MP04-A3F1-26-0007

Beekeeper

Sample Apiary Cooperative

Hive

HIVE-007, colony healthy 28 of 30 days

Harvested

18 Oct 2026

Lab report

Attached, linked to batch hash

Anchored

19 Oct 2026, tx 0x9f3...c21



Hive weight over the flow season. The small dip is a rainy day.

Sample data for illustration. Names, dates, price and transaction ID are placeholders.

Five ways the shop earns for the apiary

- **Verified price premium.** The batch page is the reason a buyer pays more than for anonymous honey.
- **Pre-orders.** The harvest forecast turns into “expected” stock. Cash arrives before extraction, and price risk drops.
- **Adopt-a-hive subscriptions.** A household reserves honey from one apiary each harvest and sees that hive's weight curve.
- **Recall and re-test.** If a lab flags a batch, only that batch is pulled. The brand survives.
- **Bulk requests.** Food makers and exporters ask for traceability by batch. The pod's records answer that directly.

Prices in the sample are placeholders. Premium micro-brands in India already list single-origin honey at ₹1,400 to ₹4,000 per kg; the cost model below uses a much more cautious figure.

Where to sell

Channel	Good for	What it asks of the seller
Own storefront	Full control of the story, subscriptions, the highest margin.	Marketing and fulfilment. A Honey Chain widget on the product page carries the verification.
Large marketplaces	Reach and trust in the checkout.	Food-safety licence, brand registration, fees. Export the batch data into the catalogue and link the verification page.
ONDC and similar open networks	Smaller sellers reaching many buyer apps.	A catalogue in the network's standard format. Onboarding rules change, so check them first.
Bulk and institutional buyers	Volume, repeat orders, exporters who need traceability.	Lab records, residue and adulteration tests, batch-level paperwork.

Who benefits

Beekeeper

- Sells verified honey above commodity price.
- Gets cash earlier through pre-orders.

- Builds a repeat customer base tied to their own hives.

Marketplace

- Fewer authenticity complaints and returns.
- A trust badge backed by records, not by a claim.
- Batch-level recall instead of delisting a whole seller.

Consumer

- Sees origin, harvest date and lab evidence before buying.
- Can re-check the same jar by scanning it at home.
- Can reorder from the same hive.

Cost of one pod against what ten hives yield

All rupee figures below are my estimates from typical Indian hobby-electronics pricing and published yield ranges. They are meant for planning, and every part price should be checked before you order.

Standard pod

Ten full nodes plus one hub.

₹34,000 typical

Range	₹25,900 to ₹42,500
Per hive	about ₹3,400
Running cost	₹5,000 to ₹10,000 a year
Yearly, all-in	about ₹16,000 (parts spread over 4 years)

Lite pod

Three full nodes, seven temperature probes, one hub.

₹17,400 typical

Range	₹12,700 to ₹22,200
Per hive	about ₹1,700
Running cost	₹4,000 to ₹8,300 a year
Yearly, all-in	about ₹10,500 (parts spread over 4 years)

Parts lists

Hive node, standard (₹2,030 to ₹3,250 each)	+
Pod hub (₹5,570 to ₹9,950)	+
Lite pod, seven wired probes (₹150 to ₹350 each)	+
Yearly running costs (₹5,000 to ₹10,000 for standard)	+

What ten hives produce

Scenario	Per hive a year	Ten hives	Basis
Apis cerana, stationary	10 kg	100 kg	Published ranges run from about 6 to 18 kg.
Apis mellifera, stationary	30 kg	300 kg	Commonly quoted as 25 to 40 kg with good care.
Apis mellifera, migratory	50 kg	500 kg	Reported averages of 50 to 60 kg with 4 to 5 harvests.

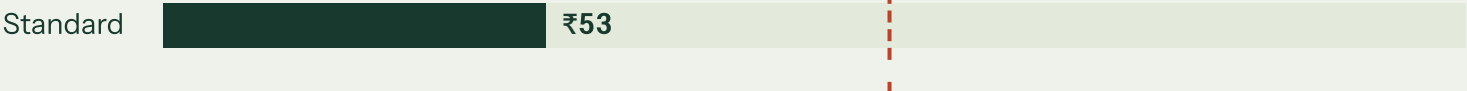
What the pod adds to every kilogram

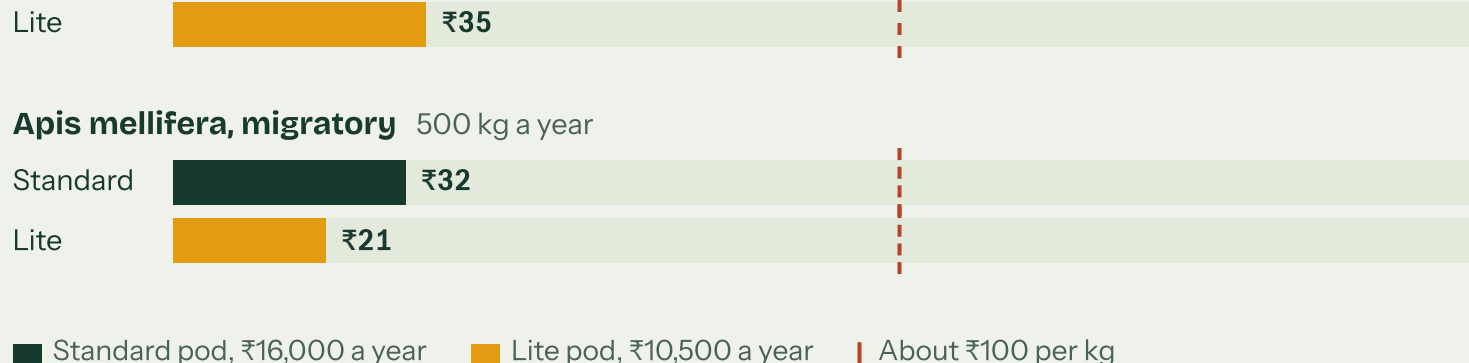
Yearly all-in pod cost divided by honey harvested. The dashed line marks about ₹100 per kg, roughly what a small beekeeper is paid at the farm gate.

Apis cerana 100 kg a year



Apis mellifera, stationary 300 kg a year





Yearly honey income for ten hives, by how it is sold

Each cell shows the income and, beneath it, what share a standard pod's ₹16,000 a year takes.

Scenario	Commodity ₹100 per kg	Cluster or bulk, verified ₹200 per kg	Direct online, net ₹450 per kg
Apis cerana, 100 kg	₹10,000 pod is 160%	₹20,000 pod is 80%	₹45,000 pod is 36%
Mellifera stationary, 300 kg	₹30,000 pod is 53%	₹60,000 pod is 27%	₹1,35,000 pod is 12%
Mellifera migratory, 500 kg	₹50,000 pod is 32%	₹1,00,000 pod is 16%	₹2,25,000 pod is 7%

Assumptions: ₹100 sits at the top of the farm-gate range small beekeepers report. The ₹200 column is my guess for what a cooperative or bulk buyer might pay for verified batches. The ₹450 column assumes a ₹700 shelf price minus about 35% for jars, labels, lab tests, courier and platform fees. Hive setup itself is separate; a ten-hive start-up is commonly put at ₹50,000 to ₹75,000.

The honest reading. At commodity prices the pod costs more than a small apiary can carry, and on 100 kg of cerana honey it costs more than the honey is worth. It starts to make sense in three cases: the honey sells verified and direct, the apiary is migratory with high volume, or the pod is shared through a cooperative.

Break-even for a standard pod on 300 kg of mellifera honey: it needs to lift the realised price by about ₹53 per kg, or add about 36 kg at the direct-sale margin (about 12% more honey), or prevent the loss of about four colonies at roughly ₹4,000 each. The Lite pod needs about ₹35 per kg. Treat the ₹4,000 colony price as an assumption and replace it with the real local figure.

Sharing it: a cooperative that owns the pods could recover cost with a lease of about ₹1,300 a month per standard pod, or about ₹900 a month per Lite pod, before any

subsidy. Check current beekeeping mission and horticulture support schemes for equipment grants.

Build it in four steps

Prove the loop cheaply first, then spend on hardware only when the software and the buyer are real.

Step	What you build	Hardware spend	How you know it worked
Hackathon demo	One real node on one hive, nine simulated hives, a laptop or phone hotspot as the hub. Batch, QR, chain anchor and consumer page working end to end.	about ₹4,000	A judge scans a QR and sees the hive's weight curve and a verified anchor.
Field pilot	One Lite pod in a real apiary for a full nectar season. Offline mode tested by switching the SIM off.	₹13,000 to ₹22,000	Sensor alerts match the beekeeper's inspection notes. No data lost across a week offline.
Cluster pilot	Five to ten pods across a cooperative, a lab partner for batch tests, and one live shop listing.	₹65,000 to ₹4,25,000	First paid orders at a verified price above commodity.
Scale	Image-based disease models, multilingual voice interface, buyer discovery, geographic analytics for institutions.	Funded by sales	Pods repay their cost from honey sales.

What this system cannot prove

- **A blockchain proves a record was not changed after anchoring.** It does not prove the honey is pure, or that the person entering the data told the truth. Purity comes from lab tests, sealed jars, sensor evidence and random audits, with the chain making those records hard to rewrite.
- **Hive weight shows nectar flow, not floral source.** A claim such as “acacia honey” needs pollen analysis or a declared and audited forage area.
- **Health thresholds are prototype rules.** Calibrate per species, region and season against beekeeper inspection notes before trusting alerts.
- **Prices here are estimates.** Component prices, SIM plans, honey prices and colony prices all move. Rebuild the tables with real quotes before presenting numbers to a funder.

Built on the Honey Chain blueprint for SIH problem statement 26021. Yield and price context drawn from the sources below; all pod costs and the ₹200 and ₹450 selling prices are estimates made for this document.

- Farmonaut, Honey farming in India: cerana and mellifera yields; ten-hive start-up cost.
- Krishi Jagran, Beekeeping in India: per-colony yield by species.
- Colony strength under stationary and migratory beekeeping, Himachal Pradesh: migratory yields.
- The Tribune, Haryana: farm-gate price and cost of production.
- Selina Wamucii, India natural honey price: wholesale benchmark.
- The Locavore: retail prices of premium single-origin honey.